



Wildlife Hazard Management Plan Miami International Airport (MIA)

14 CFR FAR Part 139.337

VOLUME III

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EXHIBITS

EXHIBIT A

MIAMI-INTERNATIONAL AIRPORT (MIA) PROPERTY BOUNDARIES & AIR OPERATIONS AREA (AOA) BOUNDARY.

EXHIBIT B

COOPERATIVE SERVICE AGREEMENT BETWEEN MIAMI-DADE AVIATION DEPARTMENT (MDAD) AND THE UNITED STATES DEPARTMENT OF AGRICULTURE ANIMAL AND PLANT HEALTH INSPECTION SERVICE (APHIS) WILDLIFE SERVICES (WS).

EXHIBIT C

FEDERAL FISH AND WILDLIFE DEPREDATION PERMIT—ISSUED BY THE DEPARTMENT OF INTERIOR, U.S. FISH AND WILDLIFE SERVICES.

EXHIBIT D

RESERVED.

1.0 INTRODUCTION

1.1 BACKGROUND

In 1999, the Miami-Dade Aviation Department (MDAD) being proactive in dealing with wildlife management contracted with Earth Tech, Inc to conduct a Wildlife Hazard Assessment (WHA) to fulfill the requirements of Title 14, Code of Federal Regulations (FAR), and Part 139.337 for Miami International Airport (MIA).

Subsequently, on June 2009 MDAD contracted with Environmental Science Associates (ESA) to conduct a review and an update of the MIA WHA and submit the Plan to the Federal Aviation Administration (FAA) for review and subsequent approval.

The Wildlife Hazard Assessment (WHA) is defined as an ecological study, conducted by a Qualified Airport Wildlife Biologist (QAWB). This Assessment analyzes local and transient wildlife populations, habitat, airport operations and strike data to establish a scientific basis for the development, implementation, and refinement of a Wildlife Hazard Management Plan (WHMP). The objective of the WHA is to provide a baseline of data and understanding of wildlife species considered hazardous on or near an airport, and attractants that provide food, water, and shelter. The assessment typically takes a year to complete.

Airports that hold Airport Certificates issued under Title 14, Code of Federal Regulations (CFR), Part 139, Certification of Airports, Subpart D, must use the standards, practices and recommendations contained in the FAA Advisory Circulars to comply with wildlife hazard management requirements.

The WHA conducted for MIA was the basis for the first Wildlife Hazard Management Plan (WHMP) that was first instituted in 2005 and the program was put into practice through the Aircraft Noise and Environmental Planning Section of the Airside Operations Division.

The approved WHMP for MIA contains MDAD's strategy for mitigating wildlife hazards that could potentially cause loss of life or otherwise jeopardize aviation operations. The WHMP is a document, designed for continual modification to address the adaptive and evolving nature of the environment, specifically species that have been identified as significant hazards to aviation either directly or indirectly. The goal is to maintain a minimal strike probability with continual monitoring, constant harassment, habitat modification and minimizing available prey. The WHMP recognizes the most significant hazards and addresses them accordingly.

The WHMP is an effective tool to assess and react to changes in wildlife populations or species, control strategies and daily changes in airfield conditions and situations.

1.2 AIRPORT SETTING AND SITE SPECIFIC

MIA is located in northeastern Miami-Dade County, Florida, approximately 6 miles northwest of downtown Miami encompassing over 3,000 acres (**Exhibit A**). The airport is designated as a passenger transport facility and serves as the primary commercial service airport and international hub in South Florida. The airport is bound on all sides by major roadways and surrounded by dense urban development to the east, north and south. To the west a mix-use commercial and residential dominates the landscape.

The MIA airfield currently consists of four air carrier runways with parallel taxiway systems. Three of the runways (8L/26R, 8R/26L and 9/27) are in an east-west configuration. The fourth runway (12/30) is a southeast-northwest diagonal runway. MIA currently provides service for 90 scheduled and 12 non-scheduled air carriers. 2018 year-end traffic reports at MIA registered a total of 416,032 operations (take-offs/landings) averaging 1,140 operations per day.

South Florida is unique in the fact that it has a tremendous seasonal variation in bird species. It is estimated that 179 species breed in Florida, and nearly 300 species occur as spring and fall migrants, or reside during the winter. The expansion or contraction of bird seasonal ranges are dynamic and can be affected by changes in regional or global weather patterns, the availability of food and water, and/or changes in other environmental factors.

Wildlife is an erratic, unpredictable aviation hazard. It varies with location, time of day, meteorological conditions, climate, human activity, geographical features and environmental changes just to mention a few. It can be unique to specific habitats, conditions and locations. The quantity and diversity of wildlife activity varies dramatically throughout North America as well as within the same region or city. Currently there is no uniform solution for controlling and mitigating this aviation hazard.

MIA turf areas are a mixture of Bahia and native grasses and wild flowers and the grounds are maintained on a three to six week mowing schedule. The mowing schedule is primarily in place in order to maintain the grass height between 6 to 11 inches which is intended to make it difficult for most birds to have a clear view of their surroundings, thus restricting their view of predators, and inhibiting movement.

Water features within MIA boundaries are limited to unnamed ditches and storm water retention ponds that serve as drainage mechanism for the airfield paved areas. Although no significant stands of emergent or submerged aquatic vegetation is present in the MIA canals, suitable edge habitat for nesting and feeding does exist intermittently along these water features.

The vast majority of MIA is devoid of trees and woody shrubs. There are areas of limited habitat which occur predominately as ornamental landscapes plantings associated with buildings, parking lots and the vehicular and pedestrian circulation routes that flow in and out of these areas. Generally these areas provide limited food and cover, and are high traffic areas that would tend to discourage any long-term use by wildlife.

2.0 DUTIES AND RESPONSIBILITIES

2.1 OVERVIEW

This section of the Wildlife Hazard Management Plan (WHMP) lists all the individuals having authority and responsibility for implementing each aspect of the plan. The section assigns or delegates specific responsibilities for various sections of the Plan to airport departments and other interested Federal, State or local agencies.

2.2 AIRSIDE DIVISION DIRECTOR OF OPERATIONS RESPONSIBILITIES:

The Division Director for Airside Operations oversees the entire Wildlife Hazard Management Plan (WHMP).

2.3 WILDLIFE MANAGEMENT SUPERVISOR (AIRSIDE SECTION CHIEF) RESPONSIBILITIES:

- a. Investigate, collect and analyze data to initiate and maintain bird and wildlife abatement to minimize conflicts with aviation operations and to update the Wildlife Hazard Management Plan (WHMP) when necessary.
- b. Acquire and maintain the necessary permits relating to bird and animal captures, relocations, and removals.
- c. Provide bird and wildlife abatement guidance and training to Airport Noise /Wildlife Officers and Airside Operations personnel.
- d. Provide bird and wildlife abatement guidance to the Maintenance Department regarding mowing and habitat modifications.
- e. Coordinate and consult with MDAD Consultant's construction and development activities to prevent and/or minimize the creation or enhancement of existing bird and wildlife attractants.
- f. Coordinate with, and provide information to, environmental consultants regarding prey base reduction initiatives and permit issues pertaining to protected species and environmentally sensitive issues.
- g. Trap, relocate, disperse, and remove birds and wildlife from airport property.
- h. Schedule Wildlife Personnel accordingly to provide adaptive response to changing bird and wildlife behaviors.
- i. Conduct accurate aircraft / wildlife strike investigation including the collection of remains for positive identification, and submittal of remains to the Smithsonian Institution, Feather Identification Lab.
- j. Inspect runway and taxiway safety areas during closures for wildlife activity including burrowing mammals.

- k. Coordinate with the United States Department of Agriculture (USDA) Animal and Plant Health Inspection Service (APHIS) Wildlife Services (WS) to trap, relocate, disperse, and remove birds & wildlife from airport property.

2.4 AIRPORT NOISE/WILDLIFE OFFICERS AND AIRSIDE OPERATIONS PERSONNEL RESPONSIBILITIES:

- a. Patrol the Air Operations Area (AOA), to include the Movement Area, from sunrise to sunset, for aviation specific bird and wildlife hazards.
- b. Trap, relocate, disperse, and remove birds and wildlife from airport property.
- c. Report significant hazardous bird and wildlife on the MDAD Wildlife Daily Observation Report.
- d. Assist the Wildlife Management Supervisor (Section Chief), and Maintenance Department with habitat modifications.
- e. Identify hazardous species and situations and report to the Wildlife Management Supervisor if corrective measures require a comprehensive lengthy process.
- f. Conduct an accurate aircraft / wildlife strike investigation including the collection of remains for positive identification, and submittal to the Smithsonian Institution, Feather Identification Lab.
- g. Inspect runway and taxiway safety areas during closures for wildlife activity including burrowing mammals.
- h. Close a runway when significant unresolved wildlife issues persist.
- i. Communicate with Air Traffic Control Tower (ATCT) and prepare NOTAMS when necessary for unusual, excessive or unresolved wildlife activity.
- j. Inspect and reopen runways when Air Traffic Control tower reports a strike.

2.5 MAINTENANCE DIVISION RESPONSIBILITIES:

- a. Mow the airfield including safety areas in a manner which minimizes attractiveness to those species most hazardous to aviation as determined by the Wildlife Management Supervisor.
- b. Maintain airfield waterways by dredging, spoil removal, and herbicide application to minimize aquatic habitats.
- c. Maintain terrestrial vegetation chemically and mechanically to prevent habitats favorable to hazardous wildlife.

- d. Maintain drainage culverts to minimize mammal incursions.
- e. Remove excessive trash from the fence line and AOA.
- f. Maintain the airport perimeter fence with a 9-foot fence constructed of 8-foot woven wire fence plus an additional foot of barbed wire for the purpose of which, besides security, is to deter wildlife from entering airport property.

2.6 THE USDA / APHIS / WS RESPONSIBILITIES:

- a. At the request of the MDAD Wildlife Management Supervisor, the USDA / APHIS WS will trap, relocate, disperse, and remove birds and wildlife from airport property.
- b. Conduct at the request of the Wildlife Supervisor annual recurring training to staff.

FAR Part 139.337(f)(2)

3.0 ATTRACTANTS / HABITAT MANAGEMENT

3.1 OVERVIEW:

Land-use practices and habitats on or near airport facilities are major factors in determining species and numbers of wildlife likely to be found in an area. All wildlife requires food, cover, and water to survive, and any habitat, which provides these basic needs, can be an attractant. By recognizing and properly managing these areas, the population of potentially hazardous wildlife species can be kept to a minimum and the likelihood of a strike incident reduced. Airport property attractants are those natural and/or manmade habitats that falls within the MIA property boundaries that may provide feeding, roosting, nesting or congregation areas for wildlife.

3.2 HAZARDS:

- a. Gulls, Great Egrets, Cattle Egrets and Turkey Vultures populations are increasing in the U.S. and are common species found to occur year round at MIA, but the potential hazard to aviation increases dramatically in the winter months of December, January, February and March when large numbers of flocks migrate to south Florida. These species are apparent indifference to aircraft activity, and they are a potentially hazardous species to aviation. Control measures will include dispersal with pyrotechnics, vehicular sirens, and removal by MDAD and/or the USDA APHIS WS.

- b. Icterids like the Boat-tailed grackle, because of their small body mass, do not normally present a significant hazard to aircraft. It is when these species are present in large flocks that the danger for a serious strike event exists. While the ingestion of one bird may not cause an engine failure, multiple ingestions involving flocks of birds would most likely cause significant damage to an aircraft.
- c. Passerines and non-passerine land birds such as the Mourning Dove and the European Starling represent a significant number of species and generally are not large enough in body mass, or congregate in numbers sufficient enough, to pose a substantial hazard to aviation at MIA.
- d. Waterfowl are not observed in significant numbers as a result of the MIA habitat is limited mainly to the drainage canals, ditches where they feed on submerged and emergent aquatic vegetation, and insects. These species are of sufficient size and mass that they would prove hazardous to aviation if a strike occurred.
- e. The Florida Burrowing Owls are listed by the State of Florida, Fish and Wildlife Conservation Commission (FWC) as a Species of Special Concern. This species are commonly found to spend much of their time observing their immediate environment while remaining near their burrow entrance. These birds usually fly at lower elevations reducing the probability of an aircraft strike. Between July 10 and February 15 of every year, MDAD will collapse inactive Borrowing owl nest burrows within the runway safety areas at MIA. Florida rule 68A-9.012 (2)(c) states that “wildlife burrows within the runway safety area as defined in 14 CFR 139.5 may be destroyed”. Hence, no State FWC permit is required for this activity.

3.3 WATER AND DRAINAGE FEATURES:

- a. At MIA local water and drainage features provide suitable habitat for herons, egrets, waterfowl and other wading and diving birds. Water features within property boundaries are limited to the unnamed ditches and storm water retention ponds that serve as drainage mechanisms for the airfield paved areas. All water featurea are inspected regularly for the above mentioned species and to make sure that no significant stands of emergent or submerged aquatic vegetation are present.

FAR Part 139.337(f)(2)(i)

4.0 MANAGEMENT PRACTICES

4.1 OVERVIEW:

Both short-term and long-term measures are in place to effectively control wildlife at MIA. These control methods are capable of providing active harassment to insure the safe operation of aircraft at MIA.

4.2 SHORT TERM MEASURES:

Short-term measures are design to meet an immediate and urgent need for dispersal of wildlife. The effectiveness of any short-term control measure is dependent on the effectiveness of the techniques and tactics that are deployed, in addition to the skill, persistence, and willingness of the airfield management and wildlife control personnel. Short-term control measures at MIA include the following:

1. Wildlife control inspections of the Airside Operations Area (AOA) sunrise to sunset all year around.
2. Documentation of all bird and wildlife activity observed during inspections.
3. Disperse birds and wildlife from the AOA whenever it is possible and safe.
4. Communicate with FAA Air Traffic Controllers to report wildlife issues.
5. Wildlife trapping and relocation.
6. Wildlife depredation by MDAD and /or USDA APHIS WS.
7. Perimeter AOA fence inspections to make sure no holes and/or gaps are available for wildlife to enter the AOA.
8. Turf height inspections and reporting.
9. Identify and report AOA pavement ponding.
10. Identify and Collapse burrows inside the runway safety areas.
11. Bird and wildlife strike investigation including the collection of biological material for submission to the Smithsonian Feather ID Lab for positive Identification.
12. Airfield mowing schedule to maintain turf at the required height.
13. Inspection of ditches and canals for wildlife and other attractants.
14. Close runways when significant unresolved wildlife issues persist.
15. Conduct annual and periodic wildlife activity reports.
16. Monitor the AOA and airport property for carrion and other wildlife attractants.
17. Inspect runway approach paths for possible bird attractants.

18. Enforce Chapter 25-2.1(g) of Miami-Dade County Rules and Regulations of No Feeding Birds and Animals at MDAD operated airports.

4.3 LONG TERM MEASURES:

Long-term control measures are designed to limit, restrict, or exclude wildlife from airport property as much as possible by reducing the local attractants that provide their basic needs for survival. These are food, water, and shelter. **The best method of providing long-term control is by managing the habitat.** While it is difficult to control all wildlife in this way due to the inherent nature of many wildlife species to adjust to changing environments, when it is used in conjunction with other methods, more successful management of wildlife at MIA can be achieved.

1. Create projects to cover all the MIA canals to reduce birds and wildlife inside the AOA.
2. Request for pavement resurfacing in areas that are known to collect standing run-off from rain events that attract birds and other wildlife to the AOA.
3. Future landscape should limit and/or restrict plant species that have seed or berries that may attract birds or other wildlife.
4. Create projects to remove isolated wooded areas or specimens that are not associated with ornamental landscaping plantings.
5. Create project to eliminate deteriorated pavement expansion joints and cracks that allow herbaceous vegetation or grasses to sprout.
6. Wildlife removal should be conducted as long-term control when other methods prove ineffective.
7. Work with the MDAD Planning Division on updating the Zoning Ordinance to protect the approach surfaces up to 10,000 feet from the runway ends with the purpose of restricting Waste Disposal Operations, Solid Waste Landfills, Composting Operations, and Recycling Centers.
8. Identify existing Solid Landfills and Recycling centers on the MIA flight paths and require from those companies a Wildlife Mitigation Plan.

5.0 HABITAT MODIFICATION

5.1 OVERVIEW:

Habitat modification is the first line of defense for limiting or restricting wildlife from an area on a long term basis and it is one of the most effective measures because it makes the airport environment less attractive to wildlife.

5.2 AIRFIELD TURF AREAS

Airfield turf areas or non-paved areas represent approximately 90% of the available habitat at MIA. The turf at MIA is generally a mixture of Bahia, native grasses, and wild flowers. Mowing cycles and herbicide applications will be ongoing throughout the year, and ground cover height is maintained between 8 and 12 inches weather permitting. The monitoring of vegetation will be ongoing throughout the year by the Maintenance Division, Airport Noise / Wildlife Officers, Airside Operations personnel, and the Wildlife Management Supervisor. Airside Operations personnel and Airport Noise / Wildlife Officers will inspect the airfield, including safety areas, for animal dens and tunneling activity and will initiate appropriate trapping or removal and will coordinate with the Maintenance Division any necessary repairs.

5.3 WATER AND DRAINAGE FEATURES

The most significant water features at MIA are the drainage canals found along the perimeter of the airfield turf areas. These canals cannot be eliminated because they are required for airfield drainage. The Maintenance Division will routinely clear emergent and submerged vegetation by cutting, dredging, or by utilizing herbicides to reduce habitat and forage. Any intermittent ponding that may form after heavy downpours, or other wet areas that occur due to seepage will be identified and filled in or eliminated by improving drainage especially if they are identified to attract wildlife.

5.4 LANDSCAPED AND WOODED AREAS

While landscape provides a limited habitat at MIA, the airport Landscape Architect will limit and restrict species that have seeds or berries that might attract birds or other wildlife. Any isolated wooded areas or individual specimens that aren't associated with ornamental landscaping plantings will be immediately removed or altered if it is determined that they provide any attraction to wildlife.

5.5 PAVED AREAS

Paved areas at MIA offer limited attraction to birds and other wildlife. An exception is during the winter months as it provides radiant heat for gulls,

pigeons and other bird species. Airport Noise / Wildlife Officers will actively patrol the airfield daily to monitor, disperse, and/or remove birds and wildlife occurring on the AOA. The Airport Noise / Wildlife Officers are trained to safely handle and discharge pyrotechnics from a variety of launchers, monitor aircraft operations to avoid the dispersal of birds or other wildlife into the approach or departure airspace, recognize and remove wildlife attractants, report birds or wildlife occurring regularly at specified locations to the Wildlife Management Supervisor, and to document bird and wildlife activity on the Wildlife Daily Observation Report.

The Maintenance Division will eliminate and remove by chemical or mechanical means herbaceous vegetation or grasses that occasionally sprout from expansion joints and cracks in the pavement; this will help reduce of species that might forage in these areas.

FAR Part 139.337(f)(2)(iii)

6.0 LAND-USE CHANGES

6.1 OVERVIEW AND RESPONSIBILITIES

The Wildlife Management Supervisor and the Airport Noise / Wildlife Officers will monitor any land-use changes that can have the potential to attract additional wildlife in the vicinity of the airport and especially within the arrival and departure corridors.

FAR Part 139.337(f)(3)

7.0 WILDLIFE CONTROL PERMITS

7.1 OVERVIEW

Wildlife is regulated at the State and Federal level. The principal agencies are the Florida Fish and Wildlife Conservation Commission (FFWCC), Florida Department of Agriculture and Consumer Services and the U.S. Fish and Wildlife Services (USFWS). The permits provide authorizations for airport personnel to harass, lethally remove, trap and relocate wildlife as well as for the application of restricted use pesticides. FFWCC consolidated its airport wildlife rules in June 2010 into a single rule, Florida Administrative Code (FAC) 68A-9.012, “Take of Wildlife on Airport Property”. The rule allows for persistent harassment on airport property of wildlife that poses safety issues for humans and aircraft without a permit, and allows for any kind of take, which includes killing wildlife in emergency safety situations. Gopher tortoises may not be killed, but must be live-captured. All wildlife burrows, such as the gopher tortoise burrows and the Florida Burrowing Owl burrows, in the Runway Safety Area (RSA) are then filled or destroyed.

Airport Noise / Wildlife Officers and Airside Operations Personnel are responsible for wildlife control on airport property. The program objective is to minimize wildlife hazards with aggressive bird and wildlife abatement. The following permits and agreements are maintained for compliance with State and Federal agencies relevant to abatement activities. These permits will be updated yearly and they will be included in the Appendix Section of this document.

7.2 LIST OF PERMITS AND AGREEMENTS

1. Cooperative Service Agreement between Miami-Dade Aviation Department (MDAD) and the United States Department of Agriculture Animal and Plant Health Inspection Service (APHIS) Wildlife Services (WS) for the capture and relocation and/or lethal shooting of mammals, birds, and other wildlife causing an immediate threat to Aviation safety. **See Exhibit “A”.**
2. Federal fish and Wildlife Depredation Permit issue by Department of Interior, U.S. Fish and Wildlife Services. **See Exhibit “B”.**

FAR PART 139.337(F)(4)

8.0 RESOURCES

8.1 PERSONNEL

MDAD will continue to maintain Airport Noise / Wildlife Officers capable of recognizing bird and wildlife hazards and appropriate abatement methods, practices, and techniques unique to the airport environment. Airside Operations personnel will continue to assist and continue to be responsible for bird and wildlife abatement as part of their assigned duties when Airport Noise / Wildlife Officers are not present on the airfield. The Maintenance Division will continue to provide personnel to conduct mowing operations and other equipment for the purpose of maintaining the airfield, modifying, and eliminating habitats that may attract hazardous wildlife.

When required, MDAD will continue to utilize and obtain the services of the USDA APHIS WS.

8.2 EQUIPMENT AND SUPPLIES

MDAD Airport Noise / Wildlife Officers will operate from vehicles equipped with:

1. VHF radio for communications with the FAA Tower.
2. 800 Mhz. radio for inter-agency communication.
3. Siren.
4. Amber light bar.

5. PA system with loud speaker.
6. Catch pole.
7. Gloves.
8. Shovel.
9. Plastic bags.
10. Eclipse Wildlife 12 Gauge Launcher.
11. Eclipse 12 gauge Bird Bomb
12. CAPA Launcher.
13. CAPA Long Range Bird Banger
14. 15 mm dispersal pistol.
15. 15 mm bangers.
16. 15 mm screamers.
17. Binoculars.
18. 12 gauge Shotgun
19. .22 & .17 Pellet rifle

8.3 OTHER EQUIPMENT MAINTAINED BY MDAD:

1. Live traps.
2. Bait for live traps.
3. Mowing equipment.
4. Tractors with trimmers and edging equipment.
5. Slope mowers.
6. M-8 Multi detonation LP Gas Cannons.
7. Sprayer equipment.
8. Pesticides

FAR PART 139.337(F)(5)

9.0 PROCEDURES TO BE FOLLOWED DURING AIR CARRIER OPERATIONS

9.1 OVERVIEW

Contained in Section (f)(5)(i) below.

10.0 ASSIGNMENT OF PERSONNEL RESPONSIBILITIES

10.1 OVERVIEW

The responsibility for the plan implementation and coordination is within the Airside Operations Division, Aircraft Noise and Environmental Planning / Wildlife Control Section, which includes the following positions and responsibilities.

10.2 PERSONNEL ASSIGNMENTS AND RESPONSIBILITIES

- a. Division Director of Airside Operations to oversee the entire program.
- b. The Wildlife Management Supervisor (Section Chief) of the Aircraft Noise and Environmental Planning / Wildlife Control Section, part of Airside Operations Division to investigate, data analysis, hazard identification and prioritization, permit maintenance, plan implementation, reporting and monitoring of bird and wildlife activity.
- c. The Airport Noise / Wildlife Officers which are part of the personnel of the Aircraft Noise and Environmental Planning / Wildlife Control Section will perform wildlife airfield patrols to detect, abate, and report wildlife activity in the movement and non-movement areas at MIA.
- d. The Airside Operations personnel (Ramp Control) are responsible for bird and wildlife abatement as part of their assigned duties when the Airport Noise / Wildlife Officers are not available.
- e. Maintenance Division personnel work in conjunction with the Airside Operations Supervisor and the Wildlife Management Supervisor (Section Chief) to minimize conditions or situations that may attract hazardous wildlife. They will be responsible for mowing, dredging, herbicide application, fence repairs, providing Lime Rock, and habitat modification if needed.
- f. Federal Aviation Administration (FAA) Air Traffic Control Tower (ATCT) will notify Airside Operations Ramp Agent in the event of a strike or significant bird or wildlife activity that requires immediate attention. Airport Noise / Wildlife Officers and Airside Operations Ramp Control personnel will respond immediately to initiate abatement, inspect runways if necessary, collect remains, and speak with the pilot if possible in order to gather accurate information for the creation of the report and submittal to the Smithsonian Institution, Feather Identification Lab.

11.0 PHYSICAL INSPECTIONS OF THE AIRCRAFT MOVEMENT AREAS CRITICAL TO SUCCESSFULLY MANAGE KNOWN WILDLIFE HAZARDS

11.1 OVERVIEW OF INSPECTIONS

Inspections will be ongoing and conducted several times daily, throughout the year. Airport Noise / Wildlife Officers will continually patrol the airfield throughout the day, from sunrise to sunset, specifically for bird and wildlife abatement and monitoring to include wildlife/bird dispersing, removal and reporting of wildlife hazards. Airside Operations Ramp Control personnel will inspect runways and taxiways several times a day and are also trained to identify, disperse, and report wildlife hazards. In the event of excessive bird or wildlife activity, Airport Noise / Wildlife Officers will coordinate with Airside Ramp personnel and the FAA ATCT to temporary modify arrival and departure air traffic until the specific hazard is remedied.

Forms and documentation of actions taken for hazards will be performed as described in Section 139.337(f)(6)(i) below.

12.0 WILDLIFE CONTROL MEASURES

12.1 OVERVIEW

Overall Wildlife control measures and procedures will be address in this section.

12.2 PROCEDURES

- a. Airport Noise / Wildlife Officers and Airside Operations Personnel are trained to:
 - 1) Safely handle and discharge pyrotechnics.
 - 2) Monitor aircraft operations to avoid dispersing birds or other wildlife into or across aircraft approach and departure corridors.
 - 3) Recognize and remove attractants occurring on and around the Airside Operations Area (AOA) from sunrise to sunset.
 - 4) Report and document bird and wildlife activity to detect recurring patterns and daily and seasonal migrations.

b. Upon observation or notification of bird or wildlife activity significant to aircraft or airport operations, Airport Noise / Wildlife Officers or Airside Operations personnel will assess the severity of the situation to determine if immediate control measure(s) should be initiated. These can include:

- 1) Harassing birds or other wildlife away from the airfield or approach and departure airspace with vehicle siren.
- 2) Harassing birds or wildlife away from the airfield or approach and departure airspace by discharging pyrotechnics.
- 3) Pending the persistence, number, and variety of species involved, multiple discharges and variety of pyrotechnics may be employed.
- 4) Harassing birds or wildlife away from the airfield or approach and departure airspace by the use of a portable and fixed propane cannons.
- 5) Each method may be used alone or/in conjunction to disperse birds and wildlife.

c. Procedures for discharging pyrotechnics:

- 1) Airport Noise / Wildlife Officers or Airside Operations personnel contacts Ramp One or Senior Agent via county radio or telephone and advise them of the wildlife hazard location, and their intentions to discharge pyrotechnics.
- 2) FAA ATCT is notified of the type and location of wildlife hazard via the ground control frequency and advised of measures that will be utilized to remove the hazard. If the hazard requires the closure of the runway for removal, the unit will contact Ramp One for coordination.
- 3) The unit verifies that no aircraft are operating within 2000' feet of area prior to discharge of pyrotechnics. Caution is taken to ensure that the wildlife is scared or forced away from any runway.
- 4) The unit notifies the FAA ATCT via ground control frequency, and Ramp One or Senior Agent, when the wildlife hazard has been removed.
- 5) Documentation is prepared.

- d. Immediate measures to alleviate wildlife hazards whenever they are detected:
- 1) Wildlife patrols will be conducted daily throughout the year by the Airport Noise / Wildlife Officers. MIA has continuous flight operations through the day; therefore wildlife patrols throughout the airfield will be conducted continuously from sunrise to sunset.
 - 2) New or immediate hazards will be minimized by early detection from daily wildlife patrols. Hazards requiring habitat modifications will be coordinated with the Maintenance Division, Airside Operations, and if required with the FAA Air Traffic Control Tower.
 - 3) Wildlife that may require immediate removal and/or relocation will be performed by the MDAD Wildlife Management Supervisor (section Chief) and/or the United States Department of Agriculture Animal and Plant Health Inspection Service (APHIS) Wildlife Services (WS). The WS is able to quickly respond as they are currently stationed at the Homestead Air Reserve Base 20 miles south of MIA.

FAR PART 139.337(F)(5)(IV)

**13.0 COMMUNICATIONS BETWEEN WILDLIFE
CONTROL PERSONNEL AND AIR TRAFFIC
CONTROL TOWER**

13.1 OVERVIEW

Anyone operating on the Movement Area will successfully complete the Movement Area Driver's Training and obtain a certificate in order to be able to drive and operate a vehicle on active taxiways, runways and areas under the control of the Air Traffic Control Tower (ATCT).

**13.2 FAR PART 139 AIRPORT FAMILIARIZATION / MOVEMENT AREA
TRAINING**

Airside Division Director, Wildlife Control Supervisor (Section Chief), Airport Noise / Wildlife Officers and Airside Operations Ramp personnel will successfully complete once every 12 consecutive calendar months recurrent FAR PART 139 Airport Familiarization / Movement Area Training.

13.3 COMMUNICATIONS WITH AIR TRAFFIC CONTROL TOWER (ATCT):

Wildlife Control Supervisor (Section Chief), Airport Noise / Wildlife Officers and Airside Operations Ramp personnel will notify the FAA ATCT via VHF radio on the appropriate frequency to advise them of the wildlife hazard location and their intentions and measures that will be utilized to remove the hazard. If the hazard requires the closure of the runway, the unit will contact Ramp One for coordination via County radio and/or phone.

FAR PART 139.337(F)(6)

14.0 PERIODIC EVALUATION AND REVIEW OF THE WILDLIFE MANAGEMENT PLAN

14.1 OVERVIEW

The Wildlife Management Supervisor will review the Wildlife Management Plan every 12 consecutive months or when any of the following events occurs:

- 1) An air carrier aircraft experiences multiple wildlife strikes;
 - 2) An air carrier aircraft experiences substantial damage from striking wildlife to an extent that the aircraft incurs any damage or structural failure that adversely affects the structural strength, performance, or flight characteristics of the aircraft and would normally require major repairs or replacement of the affected component.
 - 3) An air carrier aircraft experiences an engine ingestion of wildlife;
- Or
- 4) Wildlife of a size or in numbers capable of causing an event described in paragraphs (f)(6)(a), or (f)(6)(b), or (f)(6)(c) of this section is observed to have access the airport flight pattern or AOA.

15.0 EFFECTIVENESS IN DEALING WITH THE WILDLIFE HAZARD

15.1 OVERVIEW

The vast majority of wildlife abatement activity is the active hazing, harassment and dispersals of birds and other wildlife from sunrise to sunset every day of the year. The results of this particular activity are exceptional for controlling and removing birds and other wildlife from the Airside Operations Area (AOA), and specifically from all the runways at MIA. The constant wildlife control patrols dedicated for this purpose prevents reoccurrence of the wildlife at the same locations.

Lethal removals are immediately effective for the obvious reasons, but also provides reinforcement for the reminder population on the same flock of birds to avoid being removed.

15.2 ABATEMENT ACTIVITY

- a. Bird and Wildlife observations will be recorded daily by the Airport Noise / Wildlife Officer in the Wildlife Daily Observation Report. Each Airport Noise / Wildlife Officer is required to complete one report at the end of their shift. The form documents the date, time, wildlife type, quantity, location, dispersing technique, and what type of pyrotechnic and quantity were used. Annotations are also made to a map of the airfield depicting the exact location of the wildlife. This data accurately depicts what species are occurring on and around runways (airfield).
- b. In the event of a bird strike and if possible, the Airport Noise / Wildlife Officer will meet the aircraft at its parking location to interview the pilot. The information provided from the pilot and information provided by the FAA ATCT and Airside Operations Ramp Control personnel will be used to complete an Incident / Bird Strike Report to be submitted for review to the Wildlife Management Supervisor. The Wildlife Management Supervisor will gather additional information if possible utilizing the Airport's Noise and Operations Monitoring System (this system allows for the airport to know precisely where the bird strike occurred based on the aircraft altitude if provided by the pilot) and submit a Strike Report to the FAA. If feathers are collected, they will be submitted to the Smithsonian Institution, Feather Identification Lab once a Bird Strike is submitted electronically to the FAA and a confirmation number is obtained. All the data will accurately help depict what species are occurring on or around runways and airfield and provide a synopsis and basis for the annual evaluation.

FAR PART 139.337(F)(6)(II)

16.0 INDICATIONS THAT THE EXISTENCE OF WILDLIFE HAZARD AS PREVIOUSLY DESCRIBED IN ECOLOGICAL STUDY SHOULD BE RE-EVALUATED.

16.1 HAZARD REVIEW

Annual wildlife review will be conducted to determine significant shifts in bird and wildlife activity. Significant increases in seasonal and annual populations, strikes, or remains found, will initiate a review of previous ecological studies.

FAR PART 139.337(F)(7)

17.0 TRAINING PROGRAMS FOR PERSONNEL

17.1 WILDLIFE TRAINING

Airport Noise / Wildlife Officers and Airside Operations Ramp Control personnel are trained to include, but not limited to, the use of all pyrotechnics, firearm equipment, propane cannons, Movement Area Drivers Training, wildlife remains collection, and bird identification. Refresher training is provided every 12 consecutive calendar months. This is in addition to on-going staff education program (seminars and conferences).

The Wildlife Supervisor will attend the North America Bird Strike Committee Conference or equivalent training every year.

FAR PART 139.337(G)

18.0 FAA ADVISORY CIRCULARS FOR WILDLIFE HAZARD MANAGEMENT AT AIRPORTS

18.1 OVERVIEW

Miami-Dade Aviation Department will adhere to and consider guidance from Advisory Circulars pertaining to wildlife when feasible. This guidance will also be considered prior to and during construction projects, renovations, and other operational activities. Deviations from this guidance will be accompanied by documentation and / or data demonstrating the negligible consequences.

Miami International Airport EXHIBIT "A"



MIA—WILDLIFE HAZARD MANAGEMENT PLAN
Original Date: April 19, 2005 – Revision Date: April 2022

EXHIBIT "B"

(7/2019 version)

Agreement No.: 21-7212-8035-RA

WBS: AP.RA.RX12.72.0085

COOPERATIVE SERVICE AGREEMENT
between
MIAMI-DADE AVIATION DEPARTMENT (COOPERATOR)
and
UNITED STATES DEPARTMENT OF AGRICULTURE
ANIMAL AND PLANT HEALTH INSPECTION SERVICE (APHIS)
WILDLIFE SERVICES (WS)

ARTICLE 1 – PURPOSE

The purpose of this Cooperative Service Agreement is to cooperate in a wildlife damage management project as described in the attached Work Plan, which is attached hereto and made a part hereof.

ARTICLE 2 – AUTHORITY

APHIS-WS has statutory authority under the Acts of March 2, 1931, 46 Stat. 1468-69, 7 U.S.C. §§ 8351-8352, as amended, and December 22, 1987, Public Law No. 100-202, § 101(k), 101 Stat. 1329-331, 7 U.S.C. § 8353, to cooperate with States, local jurisdictions, individuals, public and private agencies, organizations, and institutions while conducting a program of wildlife services involving mammal and bird species that are reservoirs for zoonotic diseases, or animal species that are injurious and/or a nuisance to, among other things, agriculture, horticulture, forestry, animal husbandry, wildlife, and human health and safety.

ARTICLE 3 - MUTUAL RESPONSIBILITIES

The cooperating parties mutually understand and agree to/that:

1. APHIS-WS shall perform services set forth in the Work Plan, which is attached hereto and made a part hereof. The parties may mutually agree in writing, at any time during the term of this agreement, to amend, modify, add or delete services from the Work Plan.
2. The Cooperator certifies that APHIS-WS has advised the Cooperator there may be private sector service providers available to provide wildlife damage management (WDM) services that the Cooperator is seeking from APHIS-WS.
3. There will be no equipment with a procurement price of \$5,000 or more per unit purchased directly with funds from the cooperator for use on this project. All other equipment purchased for the program is and will remain the property of APHIS-WS.

EXHIBIT "B"

4. The cooperating parties agree to coordinate with each other before responding to media requests on work associated with this project.

ARTICLE 4 - COOPERATOR RESPONSIBILITIES

Cooperator agrees:

1. To designate the following as the authorized representative who shall be responsible for collaboratively administering the activities conducted in this agreement;

Ralph Cutié, Interim Aviation Director
Miami-Dade Aviation Department
PO Box 025504 Miami, FL 33102
(305) 342-0559
Director@miami-airport.com

2. To authorize APHIS-WS to conduct direct control activities as defined in the Work Plan. APHIS-WS will be considered an invitee on the lands controlled by the Cooperator. Cooperator will be required to exercise reasonable care to warn APHIS-WS as to dangerous conditions or activities in the project areas.
3. To reimburse APHIS-WS for costs, not to exceed the annually approved amount specified in the Financial Plan. If costs are projected to exceed the amount reflected in the Financial Plan, the agreement with amended Work Plan and Financial Plan shall be formally revised and signed by both parties before services resulting in additional costs are performed. The Cooperator agrees to pay all costs of services submitted via an invoice from APHIS-WS within 30 days of the date of the submitted invoice(s). Late payments are subject to interest, penalties, and administrative charges and costs as set forth under the Debt Collection Improvement Act of 1996.
4. To provide a Tax Identification Number or Social Security Number in compliance with the Debt Collection Improvement Act of 1996.
5. As a condition of this agreement, the Cooperator ensures and certifies that it is not currently debarred or suspended and is free of delinquent Federal debt.
6. To notify APHIS-WS verbally or in writing as far in advance as practical of the date and time of any proposed meeting related to the program.
7. The Cooperator acknowledges that APHIS-WS shall be responsible for administration of APHIS-WS activities and supervision of APHIS-WS personnel.
8. To obtain the appropriate permits for removal activities for species listed in the Work Plan and list USDA, APHIS, Wildlife Services as subpermittees.

EXHIBIT "B"

ARTICLE 5 – APHIS-WS RESPONSIBILITIES

APHIS-WS Agrees:

1. To designate the following as the APHIS-WS authorized representative who shall be responsible for collaboratively administering the activities conducted in this agreement.

Parker Hall, State Director
2820 East University Avenue
Gainesville, FL 32641
(352) 377-5556
parker.hall@usda.gov

2. To conduct activities at sites designated by Cooperator as described in the Work and Financial Plans. APHIS-WS will provide qualified personnel and other resources necessary to implement the approved WDM activities delineated in the Work Plan and Financial Plan of this agreement.
3. That the performance of wildlife damage management actions by APHIS-WS under this agreement is contingent upon a determination by APHIS-WS that such actions are in compliance with the National Environmental Policy Act, Endangered Species Act, and any other applicable federal statutes. APHIS-WS will not make a final decision to conduct requested wildlife damage management actions until it has made the determination of such compliance.
4. To invoice Cooperator quarterly for actual costs incurred by APHIS-WS during the performance of services agreed upon and specified in the Work Plan. Authorized auditing representatives of the Cooperator shall be accorded reasonable opportunity to inspect the accounts and records of APHIS-WS pertaining to such claims for reimbursement to the extent permitted by Federal law and regulations.

ARTICLE 6 – CONTINGENCY STATEMENT

This agreement is contingent upon the passage by Congress of an appropriation from which expenditures may be legally met and shall not obligate APHIS-WS upon failure of Congress to so appropriate. This agreement may also be reduced or terminated if Congress only provides APHIS-WS funds for a finite period under a Continuing Resolution.

ARTICLE 7 – NON-EXCLUSIVE SERVICE CLAUSE

Nothing in this agreement shall prevent APHIS-WS from entering into separate agreements with any other organization or individual for the purpose of providing wildlife damage management services exclusive of those provided for under this agreement.

EXHIBIT "B"

ARTICLE 8 – CONGRESSIONAL RESTRICTIONS

Pursuant to Section 22, Title 41, United States Code, no member of or delegate to Congress shall be admitted to any share or part of this agreement or to any benefit to arise therefrom.

ARTICLE 9 – LAWS AND REGULATIONS

This agreement is not a procurement contract (31 U.S.C. 6303), nor is it considered a grant (31 U.S.C. 6304). In this agreement, APHIS-WS provides goods or services on a cost recovery basis to nonfederal recipients, in accordance with all applicable laws, regulations and policies.

ARTICLE 10 – LIABILITY

APHIS-WS assumes no liability for any actions or activities conducted under this agreement except to the extent that recourse or remedies are provided by Congress under the Federal Tort Claims Act (28 U.S.C. 1346(b), 2401(b), and 2671-2680).

ARTICLE 11 – NON-DISCRIMINATION CLAUSE

The United States Department of Agriculture prohibits discrimination in all its programs and activities on the basis of race, color, national origin, age, disability, and where applicable, sex, marital status, familial status, parental status, religion, sexual orientation, genetic information, political beliefs, reprisal, or because all or part of an individual's income is derived from any public assistance program. Not all prohibited bases apply to all programs.

ARTICLE 12 - DURATION, REVISIONS, EXTENSIONS, AND TERMINATIONS

This agreement shall become effective on October 1, 2021 and shall continue through September 30, 2022. This Cooperative Service Agreement may be amended by mutual agreement of the parties in writing. The Cooperator must submit a written request to extend the end date at least 10 days prior to expiration of the agreement. Also, this agreement may be terminated at any time by mutual agreement of the parties in writing, or by one party provided that party notifies the other in writing at least 60 days prior to effecting such action. Further, in the event the Cooperator does not provide necessary funds, APHIS-WS is relieved of the obligation to provide services under this agreement.

EXHIBIT "B"

In accordance with the Debt Collection Improvement Act of 1996, the Department of Treasury requires a Taxpayer Identification Number for individuals or businesses conducting business with the agency.

Cooperator's Tax ID No.: 59-6000573
APHIS-WS's Tax ID: 41-0696271

Cooperator: 
Lester Sola, Aviation Director
Miami-Dade Aviation Department
PO Box 025504
Miami, FL 33102
Date: 9/9/2020


UNITED STATES DEPARTMENT OF AGRICULTURE
ANIMAL AND PLANT HEALTH INSPECTION SERVICE
WILDLIFE SERVICES

PARKER HALL Digitally signed by PARKER HALL
Date: 2020.09.14 13:13:14 -04'00'

Parker Hall, FL State Director
USDA, APHIS, WS
2820 East University Avenue Gainesville, FL 32641
Date: _____

EXHIBIT "B"

WORK PLAN

In accordance with the Cooperative Service Agreement between the Miami-Dade Aviation Department (MDAD) and the United States Department of Agriculture (USDA), Animal and Plant Health Inspection Service (APHIS), Wildlife Services (WS), this Work Plan sets forth the objectives, activities and budget of this project during the period of this agreement.

Program Objective

The objective of this cooperative service agreement is to assist MDAD in mitigating threats to aviation safety at Miami International Airport and other MDAD controlled airfields.

Plan of Action

Wildlife Services will use firearms, pyrotechnics, cage traps, snares, and steel traps, to harass or lethally remove wildlife that pose a flight safety hazard from the AOA, hangars, and buildings. Animals lethally removed will be disposed of according to MDAD directions. MDAD will be responsible for informing security and other persons of concern of the wildlife control activities. WS employees understand that access to the AOA is restricted in accordance with Federal law and County regulations. Access to the AOA for WS personnel will be granted only in conformance with all applicable statutes, rules, regulations, and procedures.

EXHIBIT "B"

FINANCIAL PLAN

Cost Element		Full Cost
Personnel Compensation		12,112
Travel		
Vehicles		
Other Services		
Supplies and Materials		865
Equipment		
Subtotal (Direct Charges)		12,977
Pooled Job Costs	11.00%	1,427
Indirect Costs	16.15%	2,096
Aviation Flat Rate Collection		
Agreement Total		16,500
The distribution of the budget from this Financial Plan may vary as necessary to accomplish the purpose of this agreement, but may not exceed: \$16,500		

Financial Point of Contact/Billing Address:

Cooperator Name, Address, Phone Number, Email

Norman Hegedus
Miami-Dade Aviation Department
P.O. Box 025504
Miami, FL 33102
305-342-0559
NHEGEDUS@miami-airport.com

APHIS-WS State Office Name, Address, Phone Number, Email

Vielka Dyer, Budget Analyst 2820 East
University Avenue Gainesville, FL 32641
(352) 448-2137
vielka.g.dyer@usda.gov

EXHIBIT "C"



DEPREDATION AT AIRPORTS

Permit Number: MB21982A

Version Number: 2

Effective: 2022-04-01 **Expires:** 2023-03-31

Issuing Office:

Department of the Interior
U.S. FISH AND WILDLIFE SERVICE

MB Atlanta Permit Office
1875 Century Boulevard, NE
Atlanta, Georgia 30345
permitsR4MB@fws.gov
Tel: 404-679-7070

**Carmen
Simonton**

Digitally signed by

Carmen Simonton
2022-02-14 08:03:12

Chief, Migratory Bird Permit
Office, Atlanta, Georgia

Permittee:

MIAMI DADE COUNTY
ATTENTION: NORMAN HEGEDUS
POST OFFICE BOX 025504
MIAMI, FL 33102-5504

Authority: Statutes and Regulations: 16 U.S.C 703-712 50 CFR Part 13, 50 CFR 21.41

Location where authorized activity may be conducted:

Miami International Airport, 2100 NW 42 Avenue, Miami, FL 33126
Miami - Opa locka Executive Airport 14201 NW 42 Ave, Opa-locka, FL 33054
Miami Executive Airport, 12800 SW 145th Ave, Miami, FL 33186
Miami-Homestead General Aviation Airport, 28700 SW 217 Avenue, Homestead, FL 33030
Dade-Collier Training and Transition Airport, 54575 East Tamiami Trail, Ochopee, FL 34141

Reporting requirements:

You must submit a report to your Regional Migratory Bird Permit Office even if you had no activity. Report form is at www.fws.gov/forms/3-202-9.pdf.

Authorizations and Conditions:

MIA—WILDLIFE HAZARD MANAGEMENT PLAN
Original Date: April 19, 2005 – Revision Date: April 2022

FAA Use Only

EXHIBIT “C”



DEPREDAATION AT AIRPORTS

Permit Number: MB21982A

Version Number: 2

Effective: 2022-04-01 **Expires:** 2023-03-31

Last Full Review: 2021

Next Full Review: 2026

Migratory Bird Permit Office Contact: R4BirdPermits@fws.gov (<mailto:R4BirdPermits@fws.gov>)

A. To resolve or prevent threats to human safety and/or aircraft safety at airports or airfields, you are authorized to take, temporarily possess, and transport the migratory birds specified below.

(1) **Additional authorization is required for bald eagles, golden eagles, and bird species federally listed as threatened or endangered (t/e)** (<http://www.fws.gov/endangered>).

Take of Birds of Conservation Concern (BCC) requires additional notification (see A(5) below). A list of BCC species can be found at: <https://www.fws.gov/migratorybirds/pdf/management/BCC2008.pdf>

For information on the presence of bird species at your airport, we recommend using IPAC: <https://ecos.fws.gov/ipac/>

(2) Lethal take and take of nest with viable eggs, up to: 300 migratory birds.

(3) Emergency Take:

You are authorized to exceed the take authorized above in emergency situations and/or to take of birds listed in the 2008 Birds of Conservation Concern. You must notify your Migratory Bird Permit Office (contact information above Condition A) within 48 hours and include the following information:

- (i) Emergency situation description, including date and time
- (ii) Species and number of bird(s) taken
- (iii) Method of take

A response from the office is not expected nor required. You will be contacted only if further coordination is appropriate.

(4) To minimize the lethal take of migratory birds, you are required to continually apply non-lethal methods in conjunction with lethal control. All take must be done as part of an integrated wildlife damage management program that implements nonlethal management techniques. You may not use this authority for situations in which migratory birds are merely causing a nuisance.

(5) Do not report the following activities under your Airport Depredation permit. If activities are conducted under a Depredation Order, Conservation Order, or other regulatory authorization or permit you should conduct activities in accordance with those authorizations and reporting requirements. Canada goose nests should be taken and reported under the Resident Canada Goose registration system (<https://epermits.fws.gov/eRCGR/>).

B. **Methods.** You may use the following methods of take. The use of any of the below methods is at your discretion for each situation.

EXHIBIT “C”



DEPREDATION AT AIRPORTS

Permit Number: MB21982A

Version Number: 2

Effective: 2022-04-01 **Expires:** 2023-03-31

(1) Firearms. Shotguns must be no larger than 10-gauge and must be fired from the shoulder. You must use nontoxic shot listed in 50 CFR 20.21(j). Rifles or air rifles may be used when determined most appropriate to resolve the injurious situation. Nontoxic ammunition must be used when humane and feasible. If lead shot is used, bird remains must be disposed of in a manner that prevents introducing lead in the environment. Paint ball guns may be used to haze birds but are not an authorized firearm for take. You may not use blinds, pits, or other means of concealment, decoys, duck calls, or other devices to lure or entice migratory birds into gun range. Firearm use must be in accordance with local laws and ordinances.

(2) Lethal and/or live traps. All trapping must be under humane and healthful conditions (50 CFR 13.41).
Use of Pole Traps is prohibited.

Trap-and-euthanized birds count toward the lethal take authorized under Condition A of your permit. If birds are trapped and released, birds must be released in suitable habitat in an area where they are unlikely to pose a depredation threat. When appropriate, birds should be relocated a distance sufficient to minimize potential for return to the capture site. This permit does not authorize retaining birds in captivity longer than 24 hours. Additional state and/or tribal authorization may be required for release. The Service recommends banding or marking released raptors under a USGS Bird Banding permit prior to release.

If a bird is not appropriate for release to the wild, it may be transferred as non-releasable to an individual or entity authorized to receive live birds. Approval from your Migratory Bird Permit Office is required PRIOR to transferring birds. Transferred birds count toward the lethal take authorized under Condition A, as they are removed from the wild population. Contact your Migratory Bird Permit Office prior to placement to request authorization (contact information above Condition A).

(3) Nest Take. Viable eggs may be oiled, addled, or destroyed. Eggs must be oiled using only 100% corn oil, a substance exempt from regulation by the Environmental Protection Agency. Eggs may be addled in any humane manner (see 6 below). Nests, including viable eggs, may be destroyed by any humane method, provided they are completely destroyed and eggs and/or nests are not retained after destruction.

Report take as number of active nests (not number of eggs). Do not report inactive nests taken (nests with no viable eggs or chicks present). No federal authorization is required for the take of inactive migratory bird nests.

(4) Registered animal drugs (excluding nicarbazin), pesticides, and repellents. Must be humane and used in accordance with label instructions. Additional state and/or tribal authorization may be required for use.

(5) Falconry Abatement. Migratory birds may be killed by abatement falconry birds. Birds killed by falconry abatement count toward the lethal take authorized under Condition A. Additional state and/or tribal authorization may be required.

EXHIBIT “C”



DEPREDAATION AT AIRPORTS

Permit Number: MB21982A

Version Number: 2

Effective: 2022-04-01 **Expires:** 2023-03-31

(6) Any live birds trapped or otherwise in-hand must be in humane and healthful conditions (50 CFR 13.41). Birds euthanized must follow the American Veterinary Medical Association Guidelines on Euthanasia (<https://www.avma.org/resources-tools/avma-policies/avma-guidelines-euthanasia-animals>).

C. Sick, injured, or orphaned migratory birds. You may possess and immediately transport any birds found sick, injured, or orphaned to a federally permitted rehabilitator or licensed veterinarian for care. You do not need to report these birds (50 CFR 21.31(a)), except:

Birds injured by your activities must be humanely euthanized or transferred immediately to a federally permitted migratory bird rehabilitator or a licensed veterinarian for medical care at the permittee's expense. You must report any birds injured by your activities on your Annual Report.

D. Salvage. You are authorized to salvage and temporarily possess migratory birds found dead. Salvaged birds must be disposed of as described in Condition E below within 6 months of salvage. Before you salvage any bird killed by suspected illegal activity, you must first contact the U.S. Fish and Wildlife Service Office of Law Enforcement (OLE) for authorization to salvage that bird. See FWS OLE contact information below.

Any dead bald eagle or golden eagle salvaged must be reported within 48 hours to your local U.S. Fish and Wildlife Service Office of Law Enforcement (contact information below) and to your migratory bird permit issuing office (contact information above Condition A). After clearance from OLE, contact the National Eagle Repository at (303) 287-2110 for shipment directions of these specimens.

E. Disposition of dead migratory birds. Migratory birds, nests, or eggs taken under this permit must be disposed of by one of the following:

- (1) Donated to an individual or entity authorized by permit or regulation to receive donated birds (i.e. scientific, educational, or tribal use);
- (2) Completely destroyed in accordance with local laws and ordinances;
- (3) Retained for diagnostic or personnel training purposes;
- (4) Retained and used as effigies; or
- (4) If the species is a migratory game bird and suitable for consumption, donated to a public charity.

F. Reporting.

Immediate Notification. You must immediately notify your Migratory Bird Permit Office at the contact information above Condition A about:

- (1) Emergency Take (Condition A(3))
- (2) Salvage of eagles (Condition D)

EXHIBIT “C”



DEPREDATION AT AIRPORTS

Permit Number: MB21982A

Version Number: 2

Effective: 2022-04-01 **Expires:** 2023-03-31

Annual Report. You must submit an annual report (Form 3-202-9). You must report take by species (e.g. ring-billed gull, Canada goose) and method (e.g. kill, nest take, trap-release, trap-relocate, DRC-1339).

G. **Subpermittees.** A subpermittee is an individual to whom you have provided written authorization to conduct some or all of the permitted activities in your absence. As the permittee, you are legally responsible for ensuring that your subpermittees are adequately trained and adhere to the terms of your permit. The following subpermittees are authorized: Any other person who is (1) employed by or under contract to you for the activities specified in this permit, or (2) otherwise designated a subpermittee by you in writing, may exercise the authority of this permit.

You and any subpermittees must carry a legible paper or electronic copy of this permit and display it upon request whenever you are exercising its authority. Subpermittees must be at least 18 years of age. You are responsible for maintaining current records of who you have designated as a subpermittee, including copies of any designation letters provided to individuals not named above.

H. You and any subpermittees must comply with the below Standard Conditions. **These standard conditions are a continuation of your permit conditions and must remain with your permit.** These standard conditions are nationwide and may not be modified for individual permits.

1. All of the provisions and conditions of the governing regulations at 50 CFR part 13 and 50 CFR part 21.41 are conditions of your permit. Failure to comply with the conditions of your permit could be cause for suspension of the permit. If you have questions regarding these conditions, refer to the regulations or, if necessary, contact your migratory bird permit issuing office. For copies of the regulations and forms, or to obtain contact information for your issuing office, visit:
<http://www.fws.gov/migratorybirds/mbpermits.html>.
2. General conditions set out in Subpart B of 50 CFR 13, and specific conditions contained in Federal regulations cited above, are hereby made a part of this permit. All activities authorized herein must be carried out in accord with and for the purposes described in the application submitted. Continued validity, or renewal of this permit is subject to complete and timely compliance with all applicable conditions, including the filing of all required information and reports.
3. The validity of this permit is also conditioned upon strict observance of all applicable foreign, state, local tribal, or other federal law.
4. Valid for use by permittee named above.
5. Explosive Pest Control Devices (EPCDs) are regulated by the Bureau of Alcohol, Tobacco, Firearms, and Explosives (ATF). If you plan to use EPCDs, you require a Federal explosives permit, unless you are exempt under 27 CFR 555.141. Information and contacts may be found at www.atf.gov/explosives/howto/become-an-fel.htm.

EXHIBIT “C”



DEPREDAATION AT AIRPORTS

Permit Number: MB21982A

Version Number: 2

Effective: 2022-04-01 **Expires:** 2023-03-31

6. If you encounter a migratory bird with a Federal band issued by the U.S. Geological Survey Bird Banding Laboratory, Laurel, MD, report the band number to <http://www.reportband.gov>.
7. You are responsible for obtaining appropriate, prior, written landowner permission for activity (take or release) of any migratory birds, nests, or eggs on lands where you are not the landowner or custodian.
8. You must maintain records as required in 50 CFR 13.46 and 50 CFR 21.41. All records relating to the permitted activities must be kept at the location indicated in writing by you to the migratory bird permit issuing office.
9. Acceptance of this permit authorizes the U.S. Fish and Wildlife Service to inspect any wildlife held, and to audit or copy any permits, books, or records required to be kept by the permit and governing regulations.
10. You may not conduct the activities authorized by this permit if doing so would violate the laws of the applicable State, county, municipal or tribal government or any other applicable law.

For suspected illegal activity, immediately contact USFWS Law Enforcement 1-844-FWS-TIPS (397-8477)

<https://www.fws.gov/le/regional-law-enforcement-offices.html>